

The Semicon 1.0 Story

Outlay: ₹76,000 crore

Fabs & Packaging: ₹65,000 crore; of this, ₹63,000 crore allocated across 10 projects

SCL Modernisation: ₹10,000 crore

DLI (Design Linked Incentive): ₹1,000 crore; 22 companies to get support

The government has nearly exhausted the allocation and is working on Semicon 2.0

SOURCE MEITY

Chipping Away

India launched the first phase of the ₹76,000 crore semiconductor mission in December 2021

The initial breakthroughs, though, have been on the back-end side. US memory giant Micron has committed ₹22,516 crore for an ATMP plant in Sanand

At the front-end, the Tata Group is building a wafer fabrication plant, which will soon start making chips of 28nm and above; global cutting-edge chips are 2-3nm

But leading players such as TSMC, Samsung, GlobalFoundries, and Intel continue to direct multi-billion-dollar investments into the US, Japan, and Europe

INDUSTRY • SEMICONDUCTORS



SANAND, 40 KMS from Ahmedabad airport, is home to India's semiconductor ambitions. Barely 16 months after the government approved the Murugappa Group's plan to set up a chip-packaging plant, its subsidiary, CG Semi, is on the cusp of rolling out the first commercial 'India packaged' chip under the India Semiconductor Mission (ISM). "The pilot line is important. Once the chips are qualified, it becomes easy for commercial plants to start production. This is a major milestone in our country's semiconductor journey," Ashwini Vaishnaw, Union Minister for Electronics & Information Technology, said at the inauguration of CG Semi's facility.

A few kms away, work on Micron's assembly, testing, marking, and packaging (ATMP) facility, and Kaynes Technology's outsourced semiconductor assembly and test (OSAT) plant, is advancing at a fast pace. ATMP and OSAT are back-end facilities for final stages of chip production. ATMPs are in-house while OSATs are external service providers.

Sanand, known for automobile factories, is becoming central to India's most determined push yet to build semiconductors. Its trajectory mirrors other global hotspots. In Arizona in the US, TSMC and Intel have turned the desert into a magnet for chip investments, and in Japan's Kumamoto, TSMC, Sony and Denso are turning the region into a 'silicon island'. With back-end facilities taking shape and front-end fabs in the pipeline, India, too, is building a chip-making ecosystem at a fast pace. For instance, the Tata group, working on India's first wafer fabrication plant, will soon start making chips of 28nm and above. The government is targeting 7nm and above in the next five-seven years. Globally, the cutting-edge chips are 2-3nm. A fab, or fabrication facility, is a highly sophisticated factory where semiconductor chips are made using complex processes in a controlled environment.

A STRATEGIC SHIFT

The timing couldn't be more critical. Covid-19 laid bare the vulnerabilities of global chip supply chains. The shortages, which hit almost every major industry, prompted countries to focus on indigenising production. The US rolled out the \$53 billion CHIPS and Science Act, Europe unveiled its chip ecosystem plan, China kept pumping billions into its Big Fund despite sanctions, and Taiwan, Japan, and South Korea continued treating the chip industry as national assets. India launched the first phase of the ₹76,000